

EE5750 Quantum Error Correction

Homework 1

Due: 06 Aug

As part of the Homework assessment provide solutions to only the questions marked as **(*Homework*)**. The rest of the questions are practice problems.

Linear Algebra Review

1. A collection of vectors $\{|\psi_1\rangle, \dots, |\psi_\ell\rangle\}$ is orthonormal if $\langle\psi_i|\psi_j\rangle = 0$ for $i \neq j$ and equal to 1 when $i = j$. Verify that the following sets of vectors are orthonormal.

(a) $\{|0\rangle, |1\rangle\}$

(b) $\{|+\rangle, |-\rangle\}$ where

$$|+\rangle \triangleq \frac{|0\rangle + |1\rangle}{\sqrt{2}} \text{ and } |-\rangle \triangleq \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

(c) $\{|+i\rangle, |-i\rangle\}$ where

$$|+i\rangle \triangleq \frac{|0\rangle + i|1\rangle}{\sqrt{2}} \text{ and } |-i\rangle \triangleq \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

2. *Properties of the inner product.* Consider vectors in the finite-dimensional vector space $\mathbb{C}^d$.

(a) What is the relation between $\langle\psi|\phi\rangle$ and $\langle\phi|\psi\rangle$? When are these two inner products equal?

(b) Are the following relations true? If not, what are the correct relation(s)? Here $\alpha_i \in \mathbb{C}$.

$$\langle\psi|(\alpha_1|\phi_1\rangle + \alpha_2|\phi_2\rangle) = \sum_i \alpha_i \langle\psi|\phi_i\rangle$$

$$(\alpha_1|\phi_1\rangle + \alpha_2|\phi_2\rangle)^\dagger |\psi\rangle = \sum_i \alpha_i \langle\phi_i|\psi\rangle$$

(c) When is $\langle\psi|\psi\rangle = 0$?

3. **(*Homework*)** A collection of vectors $\{|\psi_1\rangle, \dots, |\psi_\ell\rangle\}$ is linearly independent if the following is true:

$$\sum_i \alpha_i |\psi_i\rangle = 0 \text{ only when } \alpha_1 = \dots = \alpha_\ell = 0.$$

Argue that any orthonormal collection of vectors is linearly independent.

Hint: Start with $\sum_i \alpha_i |\psi_i\rangle = 0$ and apply inner product with $|\psi_j\rangle$ on both sides.

4. **(*Homework*)** *Anti-commuting matrices and trace.*

Assume that A and B anti-commute, i.e., $AB = -BA$.

(a) Show that $\text{Tr}(AB) = 0$.

(b) Suppose A is invertible, show that $\text{Tr}(B) = 0$.

Hint: In this case $B = -A^{-1}BA$.

5. (***Homework***) *More on orthonormal basis*

- (a) If $\{|\psi_1\rangle, \dots, |\psi_n\rangle\}$ is an orthonormal set of vectors in $\mathbb{C}^n$, show that for any $|\phi\rangle \in \mathbb{C}^n$ we have

$$|\phi\rangle = \sum_{i=1}^n |\psi_i\rangle \langle \psi_i | \phi \rangle,$$

which is a linear combination of $\{|\psi_i\rangle\}$.

- (b) If $\{|\psi_1\rangle, \dots, |\psi_n\rangle\}$ is any orthonormal set of vectors in $\mathbb{C}^n$, i.e., $\{|\psi_1\rangle, \dots, |\psi_n\rangle\}$ is any orthonormal basis of $\mathbb{C}^n$, argue that

$$\text{Tr}(A) = \sum_{i=1}^n \langle \psi_i | A | \psi_i \rangle.$$

6. *Properties of adjoint (conjugate transpose).*

An operator (or matrix) A is Hermitian if $A^\dagger = A$ and skew-Hermitian if $A^\dagger = -A$.

- (a) Verify $(A^\dagger)^\dagger = A$
- (b) Is $(AB)^\dagger = A^\dagger B^\dagger$?
- (c) Is $(\alpha A)^\dagger = \alpha A^\dagger$? Here $\alpha \in \mathbb{C}$.
- (d) Is $(A|\psi\rangle)^\dagger = \langle \psi | A^\dagger$?
- (e) Verify $(A + B)^\dagger = A^\dagger + B^\dagger$.
- (f) Suppose A is Hermitian, what about αA ? For what choices of α is this Hermitian, and for what choices is it skew-Hermitian?
- (g) Same as above, but assume that A is skew-Hermitian.
- (h) If A and B are Hermitian and α and β are real scalars is $\alpha A + \beta B$ Hermitian?

7. (***Homework***) *Every square matrix is a linear combination of Hermitian matrices.*

Assume that A is square.

- (a) Show that $\frac{1}{2}(A + A^\dagger)$ and $\frac{1}{2}(A - A^\dagger)$ are Hermitian and skew-Hermitian respectively.
- (b) Show that A is the sum of a Hermitian operator and a skew-Hermitian operator.
- (c) Show that A is a linear combination of two Hermitian operators, that is, there exist Hermitian operators B and C and scalars $\beta, \gamma \in \mathbb{C}$ such that $A = \beta B + \gamma C$.

Remark: This shows that every operator is a linear combination of Hermitian operators.

8. *Properties of unitary operators.*

A square matrix U is unitary when $U^{-1} = U^\dagger$.

Assume that U and V are unitary. Verify if the following statements hold; if they are false in general, under what conditions do they hold true?

- (a) UV is unitary.
- (b) $U^\dagger$ is unitary.
- (c) αU is unitary, $\alpha \in \mathbb{C}$.
- (d) $\|U|\psi\rangle\| = \||\psi\rangle\|$ for all $|\psi\rangle \in \mathbb{C}^n$
- (e) If $|\psi_i\rangle = U|\phi_i\rangle$ for $i = 1, 2$, then $\langle \psi_1 | \psi_2 \rangle = \langle \phi_1 | \phi_2 \rangle$.

Remark: Unitary operators preserve norms and inner products.

9. (***Homework***) *Positive Matrices*

We use the term *positive* to mean *positive semi-definite* matrices. An $n \times n$ operator A is positive when A is Hermitian and $\langle \psi | A | \psi \rangle \geq 0$ for all $|\psi\rangle \in \mathbb{C}^n$.

Show that the density operator corresponding to any classical state (pure or mixed) is a positive operator.

Density Operators

10. (***Homework***) Find the density operators for the following random variables. Show your labelling of the rows and columns of the density operators as well.

- (a) A binary string of length 3 is generated at random in such a way that the probability of the 3-bit string is directly proportional to the number of 1's in the string.
- (b) A binary string of length 2, where each bit of the string is an independent Bernoulli(2/3) random variable.
- (c) What is the density operator of the output when an XOR operation is applied to all the bits of the register in question (a)?
- (d) Suppose we count the number of 1's in the string stored in the register in question (b), what is the state of this output?

11. (***Homework***) Let $\mathcal{E}$ be the map

$$\mathcal{E}(A) = \sum_j K_j A K_j^\dagger,$$

where K_j are arbitrary operators. Show that if A is any positive operator (whether or not it is diagonal), then $\mathcal{E}(A)$ is positive too.
